

Analyse First order or Crossover

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January 29, 2026

When $\mu' \leq R_B$, the equation only have one solution, hence the multisolution zone must be $\mu' > R_B$. (There exists one possibility that both in $\mu' \leq R_B$ and $\mu' > R_B$ has at least one solution, but this case will not happen in CEP, so we ignore) When $\mu' > R_B$, the equation becomes

$$\begin{aligned}
 R_B &= m + \frac{8}{3\pi} \frac{\alpha_{IR}}{m_G^2} \int_{\mu'^2 - R_B^2}^{\Lambda^2} \frac{R_B s^{\frac{1}{2}}}{\sqrt{s + R_B^2}} ds \\
 &= m + \frac{8}{3\pi} \frac{\alpha_{IR} R_B}{m_G^2} \left[R_B^2 \ln(\sqrt{x + R_B^2} - \sqrt{x}) + \sqrt{x(x + R_B^2)} \right] \Big|_{\mu'^2 - R_B^2}^{\Lambda^2} \\
 &= m + \frac{8}{3\pi} \frac{\alpha_{IR} R_B}{m_G^2} \left[R_B^2 \ln \frac{\sqrt{\Lambda^2 + R_B^2} - \Lambda}{\mu' - \sqrt{\mu'^2 - R_B^2}} + \Lambda \sqrt{\Lambda^2 + R_B^2} - \mu' \sqrt{\mu'^2 - R_B^2} \right]
 \end{aligned} \tag{1}$$

When the first order transition happens, we will get the picture as below:

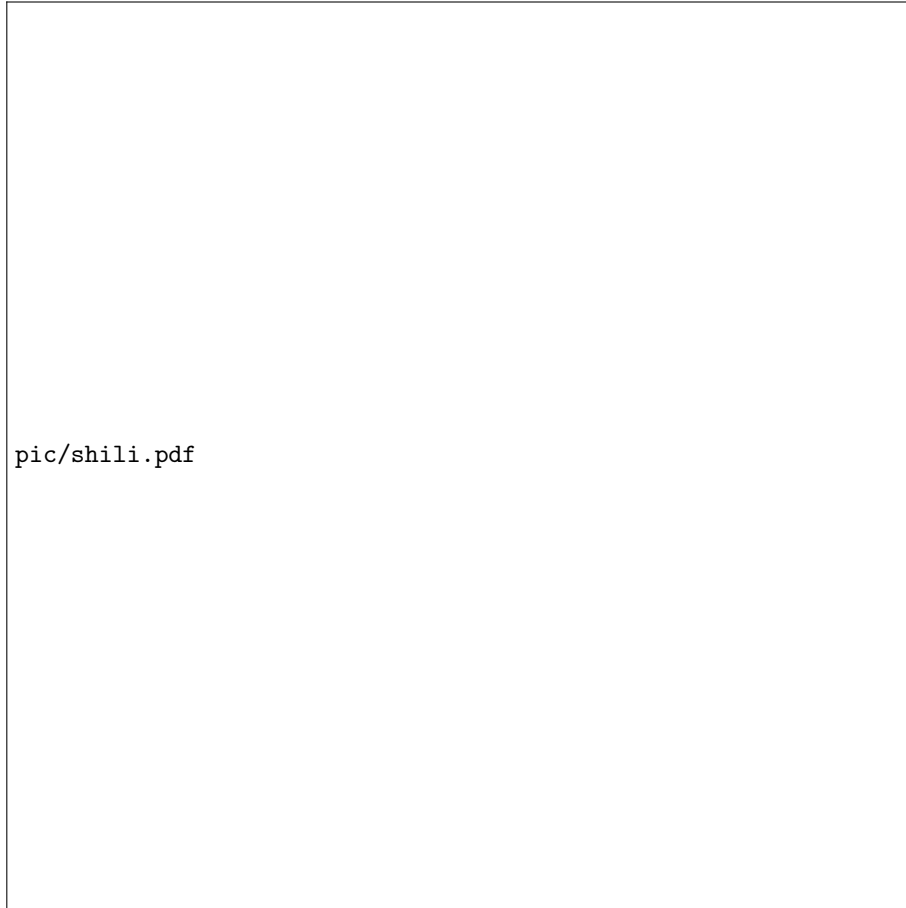


Figure 1: There exists two points $(\mu_1, R_{B1}), (\mu_2, R_{B2})$; If the transition is crossover, we could not get these two points.

Define $\frac{8}{3\pi} \frac{\alpha_{IR}}{m_G^2} = k$. At these two points, the following relationship must be satisfied:

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$$F(x) = m + kx \left[x^2 \ln \frac{\sqrt{\Lambda^2 + x^2} - \Lambda}{\mu' - \sqrt{\mu'^2 - x^2}} + \Lambda \sqrt{\Lambda^2 + x^2} - \mu' \sqrt{\mu'^2 - x^2} \right] - x = 0 \tag{2}$$

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$$F'(x) = \frac{k\Lambda(3x^2 + \Lambda^2)}{\sqrt{x^2 + \Lambda^2}} - k\mu'\sqrt{\mu'^2 - x^2} + 3kx^2 \ln \frac{\sqrt{\Lambda^2 + x^2} - \Lambda}{\mu' - \sqrt{\mu'^2 - x^2}} - 1 = 0 \quad (3)$$

Solve the simultaneous equations and eliminate the logarithm part, we could get

$$\begin{aligned} \frac{3}{2}m - kx\mu'\sqrt{\mu'^2 - x^2} - x + \frac{kx\Lambda^3}{\sqrt{\Lambda^2 + x^2}} &= 0 \\ \Leftrightarrow \mu'\sqrt{\mu'^2 - x^2} &= \frac{3}{2} \frac{m}{kx} - \frac{1}{k} + \frac{\Lambda^3}{\sqrt{\Lambda^2 + x^2}} \equiv \frac{t(x)}{2} \\ \Leftrightarrow \mu' &= \sqrt{\frac{x^2 + \sqrt{x^4 + t^2(x)}}{2}} \end{aligned} \quad (4)$$

where $t(x) = \frac{3m}{kx} - \frac{2}{k} + \frac{2\Lambda^3}{\sqrt{\Lambda^2 + x^2}}$ Substitute this equation into (??), we could get

$$\begin{aligned} \frac{kx^2\Lambda}{\sqrt{x^2 + \Lambda^2}} - \frac{1}{2} \frac{m}{x} + kx^2 \ln \frac{\sqrt{\Lambda^2 + x^2} - \Lambda}{\mu' - \sqrt{\mu'^2 - x^2}} &= 0 \\ \Leftrightarrow \frac{kx^2\Lambda}{\sqrt{x^2 + \Lambda^2}} - \frac{1}{2} \frac{m}{x} + kx^2 \ln \frac{\sqrt{\Lambda^2 + x^2} - \Lambda}{(\sqrt{x^4 + t^2} - t)^{\frac{1}{2}}} &= 0 \\ \Leftrightarrow \frac{\Lambda}{\sqrt{x^2 + \Lambda^2}} - \frac{1}{2} \frac{m}{kx^3} + \ln(\sqrt{\Lambda^2 + x^2} - \Lambda) - \frac{1}{2} \ln(\sqrt{x^4 + t^2} - t) \\ &= \frac{\Lambda}{\sqrt{x^2 + \Lambda^2}} - \frac{1}{2} \frac{m}{kx^3} + \ln \frac{\sqrt{\Lambda^2 + x^2} - \Lambda}{(\sqrt{x^4 + t^2} - t)^{\frac{1}{2}}} = 0 \equiv F(x) \end{aligned} \quad (5)$$

If first order transition, the equation above must have two solutions; if crossover, the equation do not have any solution (in $x < M_0$).

So the CEP is:

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$$F(x) = \frac{\Lambda}{\sqrt{x^2 + \Lambda^2}} - \frac{1}{2} \frac{m}{kx^3} + \ln \frac{\sqrt{\Lambda^2 + x^2} - \Lambda}{(\sqrt{x^4 + t^2} - t)^{\frac{1}{2}}} = 0 \quad (6)$$

•

$$xF'(x) = \frac{3m}{2kx^3} + \frac{xt'}{2\sqrt{x^4 + t^2}} + \frac{\Lambda^3}{(x^2 + \Lambda^2)^{\frac{3}{2}}} - \frac{t}{\sqrt{x^4 + t^2}} = 0 \quad (7)$$

$$\text{where } t' = \frac{dt}{dx} = -\frac{3m}{kx^2} - \frac{2x\Lambda^3}{(x^2 + \Lambda^2)^{\frac{3}{2}}}.$$

We have use a fact that if $F(x) = 0, F'(x) = 0$, then $x^n F(x) = 0, x^n F'(x) = 0$.

Solve the ??, we could get

$$\frac{3m}{2kx^3} + \frac{1}{\sqrt{x^4 + t^2}} \left(\frac{xt'}{2} - t \right) + \frac{\Lambda^3}{(x^2 + \Lambda^2)^{\frac{3}{2}}} = 0 \quad (8)$$

This equation is not a simple equation, we could numerically get a solution $x_0 = x(m, k, \Lambda)$, and take into ??. If $F(x_0) > 0$, first order transition; If $F(x_0) \leq 0$, crossover.